

# NanoGT Match Report: Gaucher disease

**Disease:** Gaucher disease (ORPHA:355)

**Primary gene:** GBA1

**Gene CDS:** 1491 bp

**Inheritance:** Autosomal recessive

**Target tissues scored:** hematopoietic, liver, CNS

---

## Interpretation

- At least one high-confidence precedent was found, but this is still a precedent match rather than a clinical-trial recommendation.
- Main review flags: DISEASE HETEROGENEITY: Gaucher disease has non-neuronopathic (type 1) and neuronopathic (types 2/3) subtypes requiring fundamentally different GT strategies. Specify subtype before applying these scores.; Multi-system disease; define a primary therapeutic target tissue before selecting route/vector; Vector does not naturally cover all annotated disease tissues: cns, liver.

## Disease Mechanism Evidence

**Molecular mechanism:** loss of function

**Mechanistic detail:** Beta-glucocerebrosidase lysosomal enzyme deficiency

**Gene-addition compatibility:** conditional

**Preferred modality class:** gene addition or cross correction

**Evidence level/status:** direct / source\_linked\_needs\_review

**Evidence summary:** GBA deficiency supports replacement logic but neuronopathic disease and tissue access can limit simple systemic correction

**Evidence source:** [OMIM GBA gene entry](#)

## Study-Level Limitations

- Catalog-relative ranking: current catalog contains 21 precedent programs and 8 vectors, so absence of a strong match is not proof that no therapy is possible.
  - Modality coverage is limited mainly to AAV and integrating ex vivo HSC vector precedents; dual-AAV, LNP/mRNA, genome editing, ASO, and transplant-enabling strategies are not fully represented.
  - Endpoint readiness: liver/metabolic targets may have biochemical biomarkers, but biomarker correction must be linked to clinical benefit.
  - Endpoint risk: CNS/neurodevelopmental outcomes may require natural-history data, age-stratified endpoints, and long follow-up because short-term clinical change can be hard to interpret.
  - Endpoint risk: multi-system disease may need a hierarchy of primary and secondary endpoints; one tissue response may not equal whole-disease benefit.
- 

## Top 5 GT Precedent Matches

| Rank | Program   | Vector | Score  | Confidence | Approval |
|------|-----------|--------|--------|------------|----------|
| 1    | Libmeldy  | LV     | 9.0/10 | High       | approved |
| 2    | AVR-RD-01 | LV     | 8.6/10 | High       | phase1/2 |
| 3    | ST-920    | AAV2/6 | 8.1/10 | High       | phase1/2 |
| 4    | ABO-101   | AAV9   | 8.1/10 | High       | phase1/2 |

| Rank | Program | Vector | Score  | Confidence | Approval |
|------|---------|--------|--------|------------|----------|
| 5    | RGX-121 | AAV9   | 8.1/10 | High       | phase3   |

## Match #1: Libmeldy

**Precedent disease:** Metachromatic leukodystrophy

**Vector:** LV

**Tissue target:** hematopoietic/CNS

**Composite score:** 9.0 / 10

## Score Breakdown

| Dimension                 | Score       | Max         | What it measures                                             |
|---------------------------|-------------|-------------|--------------------------------------------------------------|
| Packaging fit             | 2.00        | 2.0         | Gene CDS size vs vector cargo capacity                       |
| Tissue tropism            | 2.00        | 2.0         | Vector naturally reaches disease target tissue               |
| Protein class             | 2.00        | 2.0         | Same secreted/lysosomal/membrane/intracellular class         |
| Pathway similarity        | 2.00        | 2.0         | Same or related biological pathway                           |
| Modality compatibility    | 1.50        | 2.0         | Disease mechanism supports gene-addition precedent           |
| Inheritance compatibility | 1.00        | 1.0         | AR/XL loss-of-function pattern match                         |
| Approval precedent        | 1.00        | 1.0         | Regulatory approval / trial stage                            |
| Immunogenicity            | 2.00        | 2.0         | Pre-existing NAb seroprevalence for this vector              |
| Therapeutic window        | 0.50        | 2.0         | Can GT be given before irreversible damage?                  |
| Cross-correction          | 1.00        | 1.0         | Can transduced cells rescue untransduced neighbours?         |
| Immune privilege          | 0.90        | 1.0         | Immunological protection of target tissue                    |
| Promoter availability     | 1.00        | 1.0         | Validated tissue-specific promoters exist                    |
| Route of administration   | 1.00        | 1.0         | Established delivery route to target tissue                  |
| Organelle targeting       | 1.00        | 1.0         | Nuclear AAV delivery reaches correct subcellular compartment |
| <b>TOTAL (normalised)</b> | <b>9.00</b> | <b>10.0</b> | Raw sum / 21 × 10 (v2: 14 dimensions)                        |

## Rationale

- Gene CDS 1491bp / cargo 8000bp (19% utilized)
- Vector tropism plus precedent target match: cns, hematopoietic
- Both lysosomal enzymes — cross-correction via M6P receptor pathway likely
- Inheritance match (Autosomal recessive <-> AR)
- Pathway match: leukodystrophy
- Disease mechanism: loss of function — Beta-glucocerebrosidase lysosomal enzyme deficiency

- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: GBA deficiency supports replacement logic but neuronopathic disease and tissue access can limit simple systemic correction
- Mechanism source: OMIM GBA gene entry (<https://omim.org/entry/606463>)
- Approval status: approved
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Secreted lysosomal enzyme — maximum cross-correction via M6P receptor pathway; high therapeutic efficiency; substantially lower required transduction rate
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: ApoE/hAAT, TBG, transthyretin, albumin — extensively validated; used in Hemgenix, Roctavian, DTX301
- Route of administration: IV systemic — established, minimally invasive; used in all hepatic GT programs
- ORGANELLE TARGETING: COMPATIBLE — GBA1 standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

### Manual Review Flags

- DISEASE HETEROGENEITY: Gaucher disease has non-neuronopathic (type 1) and neuronopathic (types 2/3) subtypes requiring fundamentally different GT strategies. Specify subtype before applying these scores.
- Multi-system disease; define a primary therapeutic target tissue before selecting route/vector
- Vector does not naturally cover all annotated disease tissues: cns, liver
- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable

### Match #2: AVR-RD-01

**Precedent disease:** Fabry disease

**Vector:** LV

**Tissue target:** hematopoietic

**Composite score:** 8.6 / 10

### Score Breakdown

| Dimension                 | Score | Max | What it measures                                     |
|---------------------------|-------|-----|------------------------------------------------------|
| Packaging fit             | 2.00  | 2.0 | Gene CDS size vs vector cargo capacity               |
| Tissue tropism            | 2.00  | 2.0 | Vector naturally reaches disease target tissue       |
| Protein class             | 2.00  | 2.0 | Same secreted/lysosomal/membrane/intracellular class |
| Pathway similarity        | 2.00  | 2.0 | Same or related biological pathway                   |
| Modality compatibility    | 1.50  | 2.0 | Disease mechanism supports gene-addition precedent   |
| Inheritance compatibility | 0.70  | 1.0 | AR/XL loss-of-function pattern match                 |

| Dimension                 | Score       | Max         | What it measures                                             |
|---------------------------|-------------|-------------|--------------------------------------------------------------|
| Approval precedent        | 0.50        | 1.0         | Regulatory approval / trial stage                            |
| Immunogenicity            | 2.00        | 2.0         | Pre-existing NAb seroprevalence for this vector              |
| Therapeutic window        | 0.50        | 2.0         | Can GT be given before irreversible damage?                  |
| Cross-correction          | 1.00        | 1.0         | Can transduced cells rescue untransduced neighbours?         |
| Immune privilege          | 0.90        | 1.0         | Immunological protection of target tissue                    |
| Promoter availability     | 1.00        | 1.0         | Validated tissue-specific promoters exist                    |
| Route of administration   | 1.00        | 1.0         | Established delivery route to target tissue                  |
| Organelle targeting       | 1.00        | 1.0         | Nuclear AAV delivery reaches correct subcellular compartment |
| <b>TOTAL (normalised)</b> | <b>8.62</b> | <b>10.0</b> | Raw sum / 21 × 10 (v2: 14 dimensions)                        |

## Rationale

- Gene CDS 1491bp / cargo 8000bp (19% utilized)
- Vector tropism plus precedent target match: hematopoietic
- Both lysosomal enzymes — cross-correction via M6P receptor pathway likely
- LOF inheritance — compatible for gene replacement
- Pathway match: lysosomal storage
- Disease mechanism: loss of function — Beta-glucocerebrosidase lysosomal enzyme deficiency
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: GBA deficiency supports replacement logic but neuronopathic disease and tissue access can limit simple systemic correction
- Mechanism source: OMIM GBA gene entry (<https://omim.org/entry/606463>)
- Approval status: phase1/2
- Vector immunogenicity (LV): low (~2%) — most patients eligible; minimal screening burden
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Secreted lysosomal enzyme — maximum cross-correction via M6P receptor pathway; high therapeutic efficiency; substantially lower required transduction rate
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: ApoE/hAAT, TBG, transthyretin, albumin — extensively validated; used in Hemgenix, Roctavian, DTX301
- Route of administration: IV systemic — established, minimally invasive; used in all hepatic GT programs
- ORGANELLE TARGETING: COMPATIBLE — GBA1 standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

## Manual Review Flags

- DISEASE HETEROGENEITY: Gaucher disease has non-neuronopathic (type 1) and neuronopathic (types 2/3) subtypes requiring fundamentally different GT strategies. Specify subtype before applying these scores.
- Multi-system disease; define a primary therapeutic target tissue before selecting route/vector
- Vector does not naturally cover all annotated disease tissues: cns, liver

- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable

### Match #3: ST-920

**Precedent disease:** Fabry disease

**Vector:** AAV2/6

**Tissue target:** liver

**Composite score:** 8.1 / 10

#### Score Breakdown

| Dimension                 | Score       | Max         | What it measures                                             |
|---------------------------|-------------|-------------|--------------------------------------------------------------|
| Packaging fit             | 1.50        | 2.0         | Gene CDS size vs vector cargo capacity                       |
| Tissue tropism            | 2.00        | 2.0         | Vector naturally reaches disease target tissue               |
| Protein class             | 2.00        | 2.0         | Same secreted/lysosomal/membrane/intracellular class         |
| Pathway similarity        | 2.00        | 2.0         | Same or related biological pathway                           |
| Modality compatibility    | 1.50        | 2.0         | Disease mechanism supports gene-addition precedent           |
| Inheritance compatibility | 0.70        | 1.0         | AR/XL loss-of-function pattern match                         |
| Approval precedent        | 0.50        | 1.0         | Regulatory approval / trial stage                            |
| Immunogenicity            | 1.50        | 2.0         | Pre-existing NAb seroprevalence for this vector              |
| Therapeutic window        | 0.50        | 2.0         | Can GT be given before irreversible damage?                  |
| Cross-correction          | 1.00        | 1.0         | Can transduced cells rescue untransduced neighbours?         |
| Immune privilege          | 0.90        | 1.0         | Immunological protection of target tissue                    |
| Promoter availability     | 1.00        | 1.0         | Validated tissue-specific promoters exist                    |
| Route of administration   | 1.00        | 1.0         | Established delivery route to target tissue                  |
| Organelle targeting       | 1.00        | 1.0         | Nuclear AAV delivery reaches correct subcellular compartment |
| <b>TOTAL (normalised)</b> | <b>8.14</b> | <b>10.0</b> | Raw sum / 21 × 10 (v2: 14 dimensions)                        |

#### Rationale

- Gene CDS 1491bp / cargo 4700bp (32% utilized)
- Vector tropism plus precedent target match: liver
- Both lysosomal enzymes — cross-correction via M6P receptor pathway likely
- LOF inheritance — compatible for gene replacement
- Pathway match: lysosomal\_storage
- Disease mechanism: loss of function — Beta-glucocerebrosidase lysosomal enzyme deficiency
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints

- Mechanism evidence: GBA deficiency supports replacement logic but neuronopathic disease and tissue access can limit simple systemic correction
- Mechanism source: OMIM GBA gene entry (<https://omim.org/entry/606463>)
- Approval status: phase1/2
- Vector immunogenicity (AAV2/6): moderate (~17%) — significant proportion may require pre-screening or exclusion
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Secreted lysosomal enzyme — maximum cross-correction via M6P receptor pathway; high therapeutic efficiency; substantially lower required transduction rate
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: ApoE/hAAT, TBG, transthyretin, albumin — extensively validated; used in Hemgenix, Roctavian, DTX301
- Route of administration: IV systemic — established, minimally invasive; used in all hepatic GT programs
- ORGANELLE TARGETING: COMPATIBLE — GBA1 standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

### Manual Review Flags

- DISEASE HETEROGENEITY: Gaucher disease has non-neuronopathic (type 1) and neuronopathic (types 2/3) subtypes requiring fundamentally different GT strategies. Specify subtype before applying these scores.
- Multi-system disease; define a primary therapeutic target tissue before selecting route/vector
- Vector does not naturally cover all annotated disease tissues: cns, hematopoietic
- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone

### Match #4: ABO-101

**Precedent disease:** Mucopolysaccharidosis type IIIB

**Vector:** AAV9

**Tissue target:** CNS

**Composite score:** 8.1 / 10

### Score Breakdown

| Dimension              | Score | Max | What it measures                                     |
|------------------------|-------|-----|------------------------------------------------------|
| Packaging fit          | 1.50  | 2.0 | Gene CDS size vs vector cargo capacity               |
| Tissue tropism         | 2.00  | 2.0 | Vector naturally reaches disease target tissue       |
| Protein class          | 2.00  | 2.0 | Same secreted/lysosomal/membrane/intracellular class |
| Pathway similarity     | 2.00  | 2.0 | Same or related biological pathway                   |
| Modality compatibility | 1.50  | 2.0 | Disease mechanism supports gene-addition precedent   |

| Dimension                 | Score       | Max         | What it measures                                             |
|---------------------------|-------------|-------------|--------------------------------------------------------------|
| Inheritance compatibility | 1.00        | 1.0         | AR/XL loss-of-function pattern match                         |
| Approval precedent        | 0.50        | 1.0         | Regulatory approval / trial stage                            |
| Immunogenicity            | 1.00        | 2.0         | Pre-existing NAb seroprevalence for this vector              |
| Therapeutic window        | 0.50        | 2.0         | Can GT be given before irreversible damage?                  |
| Cross-correction          | 1.00        | 1.0         | Can transduced cells rescue untransduced neighbours?         |
| Immune privilege          | 0.90        | 1.0         | Immunological protection of target tissue                    |
| Promoter availability     | 1.00        | 1.0         | Validated tissue-specific promoters exist                    |
| Route of administration   | 1.00        | 1.0         | Established delivery route to target tissue                  |
| Organelle targeting       | 1.00        | 1.0         | Nuclear AAV delivery reaches correct subcellular compartment |
| <b>TOTAL (normalised)</b> | <b>8.05</b> | <b>10.0</b> | Raw sum / $21 \times 10$ (v2: 14 dimensions)                 |

## Rationale

- Gene CDS 1491bp / cargo 4700bp (32% utilized)
- Vector tropism plus precedent target match: cns
- Both lysosomal enzymes — cross-correction via M6P receptor pathway likely
- Inheritance match (Autosomal recessive <-> AR)
- Pathway match: lysosomal storage
- Disease mechanism: loss of function — Beta-glucocerebrosidase lysosomal enzyme deficiency
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: GBA deficiency supports replacement logic but neuronopathic disease and tissue access can limit simple systemic correction
- Mechanism source: OMIM GBA gene entry (<https://omim.org/entry/606463>)
- Approval status: phase1/2
- Vector immunogenicity (AAV9): high (~22%) — substantial patient exclusion expected; immunodepletion protocols may be needed
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Secreted lysosomal enzyme — maximum cross-correction via M6P receptor pathway; high therapeutic efficiency; substantially lower required transduction rate
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: ApoE/hAAT, TBG, transthyretin, albumin — extensively validated; used in Hemgenix, Roctavian, DTX301
- Route of administration: IV systemic — established, minimally invasive; used in all hepatic GT programs
- ORGANELLE TARGETING: COMPATIBLE — GBA1 standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

## Manual Review Flags

- DISEASE HETEROGENEITY: Gaucher disease has non-neuronopathic (type 1) and neuronopathic (types 2/3) subtypes requiring fundamentally different GT strategies. Specify subtype

before applying these scores.

- Multi-system disease; define a primary therapeutic target tissue before selecting route/vector
- Vector does not naturally cover all annotated disease tissues: hematopoietic
- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone

## Match #5: RGX-121

**Precedent disease:** Mucopolysaccharidosis type II

**Vector:** AAV9

**Tissue target:** CNS/liver

**Composite score:** 8.1 / 10

### Score Breakdown

| Dimension                 | Score       | Max         | What it measures                                             |
|---------------------------|-------------|-------------|--------------------------------------------------------------|
| Packaging fit             | 1.50        | 2.0         | Gene CDS size vs vector cargo capacity                       |
| Tissue tropism            | 2.00        | 2.0         | Vector naturally reaches disease target tissue               |
| Protein class             | 2.00        | 2.0         | Same secreted/lysosomal/membrane/intracellular class         |
| Pathway similarity        | 2.00        | 2.0         | Same or related biological pathway                           |
| Modality compatibility    | 1.50        | 2.0         | Disease mechanism supports gene-addition precedent           |
| Inheritance compatibility | 0.70        | 1.0         | AR/XL loss-of-function pattern match                         |
| Approval precedent        | 0.80        | 1.0         | Regulatory approval / trial stage                            |
| Immunogenicity            | 1.00        | 2.0         | Pre-existing NAb seroprevalence for this vector              |
| Therapeutic window        | 0.50        | 2.0         | Can GT be given before irreversible damage?                  |
| Cross-correction          | 1.00        | 1.0         | Can transduced cells rescue untransduced neighbours?         |
| Immune privilege          | 0.90        | 1.0         | Immunological protection of target tissue                    |
| Promoter availability     | 1.00        | 1.0         | Validated tissue-specific promoters exist                    |
| Route of administration   | 1.00        | 1.0         | Established delivery route to target tissue                  |
| Organelle targeting       | 1.00        | 1.0         | Nuclear AAV delivery reaches correct subcellular compartment |
| <b>TOTAL (normalised)</b> | <b>8.05</b> | <b>10.0</b> | Raw sum / 21 × 10 (v2: 14 dimensions)                        |

### Rationale

- Gene CDS 1491bp / cargo 4700bp (32% utilized)
- Vector tropism plus precedent target match: cns, liver
- Both lysosomal enzymes — cross-correction via M6P receptor pathway likely



- LOF inheritance — compatible for gene replacement
- Pathway match: lysosomal\_storage
- Disease mechanism: loss of function — Beta-glucocerebrosidase lysosomal enzyme deficiency
- Gene-addition modality compatibility: conditionally supports gene addition; review disease-specific constraints
- Mechanism evidence: GBA deficiency supports replacement logic but neuronopathic disease and tissue access can limit simple systemic correction
- Mechanism source: OMIM GBA gene entry (<https://omim.org/entry/606463>)
- Approval status: phase3
- Vector immunogenicity (AAV9): high (~22%) — substantial patient exclusion expected; immunodepletion protocols may be needed
- Very narrow therapeutic window — congenital or rapidly fatal early onset; in utero or immediate neonatal GT required; substantially increases trial complexity
- Secreted lysosomal enzyme — maximum cross-correction via M6P receptor pathway; high therapeutic efficiency; substantially lower required transduction rate
- Immune privilege: high privilege — blood-brain barrier severely limits T-cell access; durable expression expected
- Promoter availability: ApoE/hAAT, TBG, transthyretin, albumin — extensively validated; used in Hemgenix, Roctavian, DTX301
- Route of administration: IV systemic — established, minimally invasive; used in all hepatic GT programs
- ORGANELLE TARGETING: COMPATIBLE — GBA1 standard subcellular localisation; nuclear AAV delivery directly produces functional protein at the correct cellular compartment with no additional organelle-import steps required.

### Manual Review Flags

- DISEASE HETEROGENEITY: Gaucher disease has non-neuronopathic (type 1) and neuronopathic (types 2/3) subtypes requiring fundamentally different GT strategies. Specify subtype before applying these scores.
- Multi-system disease; define a primary therapeutic target tissue before selecting route/vector
- Vector does not naturally cover all annotated disease tissues: hematopoietic
- Narrow therapeutic window; evaluate newborn screening, presymptomatic diagnosis, or very early dosing feasibility
- Mechanism evidence is conditionally compatible with gene addition; review the listed disease-specific constraints before treating vector precedent as transferable
- AAV tropism is species- and route-dependent; confirm human target-cell biodistribution rather than relying on animal tropism alone